```
#define LED 4

void setup() {
 pinMode(LED, OUTPUT);
}
void loop() {
 digitalWrite(LED, HIGH);
 delay(500);
 digitalWrite(LED, LOW);
 delay(500);
}
```

Exp2

```
#define button_pin 16
#define led_pin 18
int button_state =0;

void setup() {
 pinMode(led_pin , OUTPUT);
 pinMode(button_pin, INPUT_PULLUP);
 Serial.begin(115200);
}
void loop()
{
 button_state=digitalRead(button_pin);
 if(button_state==LOW)
 {
  digitalWrite(led_pin, HIGH);
  Serial.println(" LED ON ");
 }
  else
```

```
 {
  digitalWrite(led_pin, LOW);
  Serial.println(" LED OFF ");
 }
}
```

Exp3 potentiometer

```
void setup() {
 Serial.begin(115200);
 pinMode(32, INPUT);
 pinMode(2, OUTPUT);
}

void loop() {
 int value=analogRead(32);
 int intensity = map(value,0,4098,0,255);
 analogWrite(2,intensity);
 Serial.println(intensity);
 delay(10);
}
```

Exp4 LDR

```
void setup()
{
 Serial.begin(9600);
 pinMode(32, OUTPUT);
 pinMode(34,INPUT);
}
```

```
void loop()
{
 int LDRStatus = analogRead(34);
 if (LDRStatus <=1000)
 {
  digitalWrite(32, LOW);
  Serial.print("Current light intensity value is - ");
  Serial.println(LDRStatus);
 }
 else
 {
  digitalWrite(32, HIGH);
  Serial.print("Current light intensity value is - ");
  Serial.println(LDRStatus);
 }

}
```

Exp 5 PIR

```
void setup()
{
 pinMode(4, INPUT);
 pinMode(16, OUTPUT);
 Serial.begin(9600);
}

void loop()
{
 int motion= 0;
```

```
 motion = digitalRead(4);
 if (motion == HIGH)
 {
  digitalWrite(16, HIGH);
  Serial.println(motion);
  Serial.println("motion detected");
 }
 else
 {
  digitalWrite(16, LOW);
  Serial.println(motion);
  Serial.println("motion  not detected");
 }
 delay(1000); // Delay a little bit to improve simulation performance
}
```

Exp 6 DHT22 (DHT sensor library)

```
#include "DHT.h"
#define DHT22PIN 26
DHT dht(DHT22PIN, DHT22);
void setup() {
 pinMode(15, OUTPUT);
 Serial.println("Hello, ESP32!");
 Serial.begin(115200);
/* Start the DHT22 Sensor */
 dht.begin();
}
void loop() {
 delay(1000); // this speeetup code heds up the simulation
 float humi = dht.readHumidity();
```

```
 float temp = dht.readTemperature();
 Serial.print("Temperature: ");
 Serial.print(temp);
 Serial.print("ºC ");
 Serial.print("Humidity: ");
 Serial.println(humi);
 if(temp > 30){
  digitalWrite(15, HIGH);
 }else if(temp < 25){
  digitalWrite(15, LOW);
 }
 delay(1000);
}
```

Exp 7 ultrasonic

```
float distance;
float duration;
void setup() {
 Serial.begin(9600);
 // trig pin(triggers the ultrasonic sensor to throw waves)
 pinMode(18, OUTPUT);
 //echo pin(a singal is sent to the arduino if any obstruction is found in the waves)
 pinMode(17, INPUT);
 pinMode(4, OUTPUT);
}
void loop() {
 digitalWrite(18, HIGH);
 delay(10); // wait for 10 microseconds
 digitalWrite(18, LOW);
 // calculates the duration of the wave after obstruction
 duration = pulseIn(17, HIGH);
```

```
 // calculates the distance between the object that obstructed and the sensor
 distance = duration * 0.017;
 if(distance < 50)
 {
  digitalWrite(4, HIGH);
 }
 else
 {
  digitalWrite(4, LOW);
 }
 Serial.print("distance: ");
 Serial.print(distance );
 Serial.print(" cm");


}
```

Exp 8 gas sessor

```
void setup() {
 Serial.begin(115200);
 pinMode(2, OUTPUT);
}
void loop() {
 int gasvalue = analogRead(4);
 Serial.print("Gas Sensor Value: ");
 Serial.print(gasvalue);
 if(gasvalue<=700){
  digitalWrite(2, HIGH);
  Serial.println("Danger ! Gas leak Detected!");
 }
 else{
```

```
  digitalWrite(2, LOW);
  Serial.println("Environment safe");
 }
 delay(2000);
}
```

Exp9 servomoter

(Servo

ESP32Servo)

```
#include <ESP32Servo.h>
#define SERVO_PIN 19 // ESP32 pin GPIO26 connected to servo motor
Servo servoMotor;
void setup() {
 servoMotor.attach(SERVO_PIN);  // attaches the servo on ESP32 pin
}
void loop() {
 // rotates from 0 degrees to 180 degrees
 for (int pos = 0; pos <= 180; pos += 1) {
  // in steps of 1 degree
  servoMotor.write(pos);
  delay(15); // waits 15ms to reach the position
 }
 // rotates from 180 degrees to 0 degrees
 for (int pos = 180; pos >= 0; pos -= 1) {
  servoMotor.write(pos);
  delay(15); // waits 15ms to reach the position
 }
}
```